

The Perturbation Catalogue:

An Integrated Resource for Exploring the Impact of Genetic Perturbations

ECCB 2026 / GENEVA
31 AUG--4 SEP

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Every perturbation experiment asks what changes when a gene—or one amino acid—changes. The Catalogue turns scattered screens into a navigable evidence landscape: rigorously curated, normalised to Ensembl, aligned to ontologies, and increasingly reprocessed through shared pipelines.

1,201

CRISPR SCREENS

Gene-level fitness and phenotype scores across loss- and gain-of-function screens.

26

PERTURB-SEQ

Single-cell expression responses with standardised differential expression and pathway enrichment.

10

MAVE

Variant-level functional effects from deep mutational scanning.

A COMMON EVIDENCE LANGUAGE

study • perturbation • assay • model system • target • provenance • licence

CURATE / ONTOLOGIES / ENSEMBL IDS / REPROCESS / ANALYSE / SERVE

RAW READS BECOME PUBLIC EVIDENCE

PERTURB-SEQ REPROCESSING / FIVE DATASETS LIVE



A shared workflow reduces avoidable study-to-study variation while preserving explicit provenance. The first production set spans Nadig 2025 (Jurkat, HepG2) and Replogle 2022 (K562 essential, K562 genome-wide, RPE1 essential).

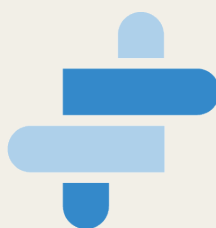
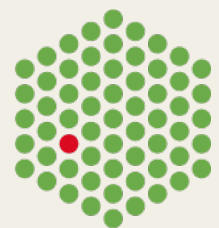
OUTPUT / differential expression / significant up + down effects / pathway enrichment / downloadable full datasets

WHAT DATA OR CAPABILITY IS YOUR WORK MISSING?

Request datasets or functionality, propose a collaboration, or fund a multi-year programme around the modalities and questions your community needs.

NEXT / AI-ASSISTED CURATION / BROADER REANALYSIS / PERIODIC RELEASES
ACCESS / PORTAL / REST API / PARQUET / CSV.GZ / METADATA JSON

EMBL-EBI



Open Targets

ebi.ac.uk/perturbation-catalogue

EXPLORE



CONTACT



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